MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

JAN 5981

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Reflex, integral cavity, mechanically tuned frequency range 1245 to 1460 Mc.

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Ik	F	tk	TA	Alt
Unit:	V	Vdc	Vdc	mAdc	Mc	(See note 1)	°C	ft
Maximum:	6.8	250	-350	50	---	---	75	50,000
Minimum:	5.8	---	-20	---	---	---	---	---

PHYSICAL CHARACTERISTICS: (See outline drawing)

TEST CONDITION:

Parameter:	Ef	Ers	Er	F
Unit:	V	Vdc	Vdc	Mc
	6.3	225	-30 to -220	F1 = 1,245 ± 0.3% F2 = 1,350 ± 0.3% F3 = 1,460 ± 0.3% (See note 2)

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required

5. Preparation for delivery

E-50.2 Holding period (t = 168 hours)

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification					
1023	Temperature coefficient	Er max Po; F = F2 mode A; T = 40°C	ΔF/°C	---	±0.05	Mc °C
1031	Low-frequency vibration	No voltage; t = 120	---	---	---	---
	Quality conformance inspection, part 1					
4102	Operation bump	Er max Po; mode A; F = F2	ΔPo/Po	---	±10	%
1256	Cathode current (1)		Ik	---	46.0	mAdc
1256	Cathode current (2)	Ef = 6.3 V and 5.8 V, F = F3, t = 1 min	ΔIk/Ik	---	5.0	mAdc
4213	Reflector voltage (1)	Er max Po; F = F1; mode A	Er	-45	-105	Vdc
4213	Reflector voltage (2)	Er max Po; F = F2; mode A	Er	-85	-145	Vdc
4213	Reflector voltage (3)	Er max Po; F = F3; mode A	Er	-145	-205	Vdc
4229	† Total reflector current	Er = -150 Vdc	Ir	---	7.0	μAdc
4229	† Reflector-leakage current	Er = -150 Vdc	Ir	---	5.0	μAdc
4229	† Reflector-gas current	Er = -150 Vdc	Ir	---	2.0	μAdc
4218	Tunable frequency Max frequency Min frequency	Er max Po	--- F F	--- 1,460 ---	--- --- 1,245	--- Mc Mc
4211	Power output (1)	F = F1; Er max Po	Po	40	---	mW
4211	Power output (2)	F = F3; Er max Po	Po	40	---	mW
	Quality conformance inspection, part 2					
1211	Insulation of electrodes	250 Vdc; tube cold	Rk-rs Rh-rs	2.0 2.0	--- ---	Meg Meg
1301	Heater current		If	425	485	mA
1336	Heater-cathode leakage	Ehk = ±45 Vdc	Ihk	---	100	μAdc
4211	Power output (3)	F = F2; Er max Po	Po	40	---	mW
4201	Electronic tuning range (1)	Er 50% max Po; F = F1	ΔF	5.0	---	Mc
4201	Electronic tuning range (2)	Er 50% max Po; F = F2	ΔF	4.0	---	Mc
4201	Electronic tuning range (3)	Er 50% max Po; F = F3	ΔF	2.5	---	Mc
4231	Hysteresis		---	---	15	%

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4.7.5.3.2 Durability testing procedure - In performing the test:

METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 2 -Continued					
---	Mechanical tuning rate	F1 through F3	Tuner turns	1-1/2	2-1/2	---
	Quality conformance inspection, part 3					
4.7	Life-test provisions	Group C; F = F3	t	500	---	hr
4.7.3	Life-test end points		---	---	---	---
4211	Power output (2)	F = F3; Er/max Po	Po	25	---	mW

NOTES:

1. No warmup time is required in order to obtain the specified life from the tube. If extended life is desired, a heater warmup period of approximately 1 minute is recommended before the application of the resonator voltage.
2. The tube shall be fixed firmly in a suitable socket by clamps which make contact at the base and adjacent flange only. The measurements on tubes in an oscillating state shall be made with the output line coupled through a connector as specified on Drawing 275-JAN into a 50 ohm resistive load.

Custodians:

Army - EL
Navy - SH
Air Force - 11

Review activities:

Army - EL
Navy - SH
Air Force - 11, 17, 85

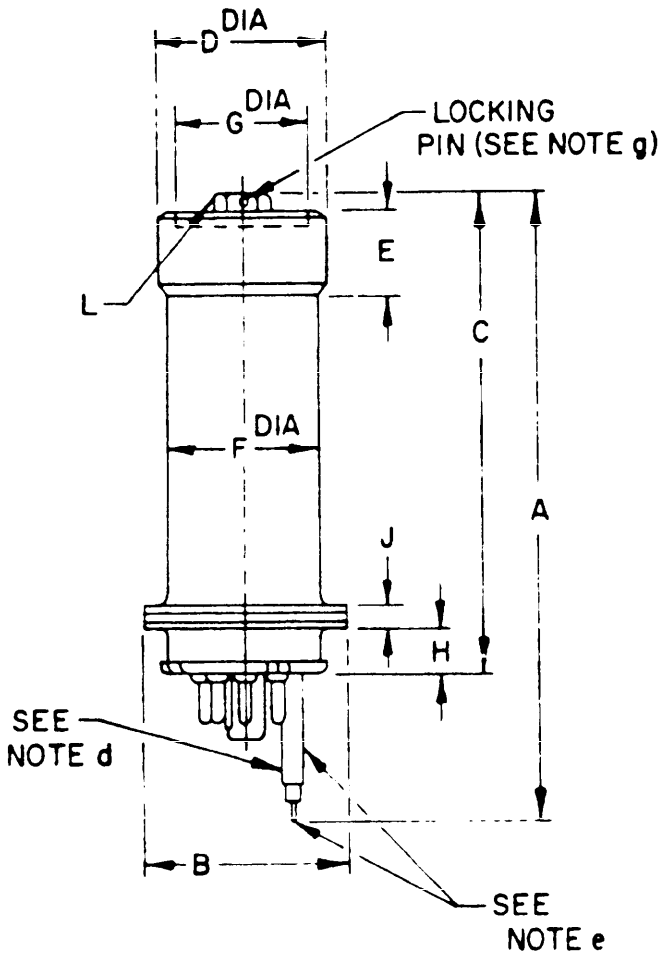
User activities:

Army - MU
Navy - WP, MC, CG
Air Force - None

Preparing activity:
Navy - SH

(Project 5960-1891)

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DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
Qualification				
D		1.375		34.93
E		.594		15.09
L	Hex nut: across flats .438(11.13)			
Quality conformance inspection, part 1				
A		5.250		133.35
B		1.656		42.06
C		4.063		103.20
Quality conformance inspection, part 2				
H	.289	.351	7.34	8.92
J		.141		3.58
Nominal dimensions				
F	1.250		31.75	
G	1.187		30.15	

Base connections	
Pin No.	Element
1	rs
2	h
3	nc
4	output lead
5	r
6	nc
7	h
8	k

NOTES:
a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
b. Nominal dimensions are for information and are not required for inspection purposes.
c. The AQL for the combined mechanical defectives in quality conformance inspection, part 1, shall be 1.0 percent, Inspection level I.
d. Nickel (30 MSI silver plate permissible).
e. Inner and outer conductors of coaxial output lead shall be concentric within .010 (.254 mm) inches.
f. For base pin alinement use gage GB8-2.
g. Tuning nut secured to tuner shaft by means of locking pin.
h. Base: B7-91

FIGURE 1. Outline drawing of electron tube type 5981.

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